

Given: Due:	Further Maths (Statistics) HW 23	Name:																					
Chapter 6: Chi-Squared tests (goodness of fit)																							
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.																							
91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)																		
<i>Useful resources:</i> Your notes from recent lessons & your text book																							
Questions - do these on lined paper																							
Q1. A spinner used for a game is designed to give scores with the following probabilities																							
<table border="1" style="float: right; margin-left: auto; margin-right: auto;"> <tr> <td style="padding: 5px;">Score</td> <td style="padding: 5px;">1</td> <td style="padding: 5px;">2</td> <td style="padding: 5px;">3</td> <td style="padding: 5px;">4</td> <td style="padding: 5px;">6</td> </tr> <tr> <td style="padding: 5px;">Probability</td> <td style="padding: 5px;">$\frac{3}{10}$</td> <td style="padding: 5px;">$\frac{1}{10}$</td> <td style="padding: 5px;">$\frac{1}{10}$</td> <td style="padding: 5px;">$\frac{2}{5}$</td> <td style="padding: 5px;">$\frac{1}{10}$</td> </tr> </table>						Score	1	2	3	4	6	Probability	$\frac{3}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{5}$	$\frac{1}{10}$						
Score	1	2	3	4	6																		
Probability	$\frac{3}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{5}$	$\frac{1}{10}$																		
The spinner is spun 80 times and the results are as follows																							
<table border="1" style="float: right; margin-left: auto; margin-right: auto;"> <tr> <td style="padding: 5px;">Score</td> <td style="padding: 5px;">1</td> <td style="padding: 5px;">2</td> <td style="padding: 5px;">3</td> <td style="padding: 5px;">4</td> <td style="padding: 5px;">6</td> </tr> <tr> <td style="padding: 5px;">Frequency</td> <td style="padding: 5px;">15</td> <td style="padding: 5px;">4</td> <td style="padding: 5px;">12</td> <td style="padding: 5px;">41</td> <td style="padding: 5px;">8</td> </tr> </table>						Score	1	2	3	4	6	Frequency	15	4	12	41	8						
Score	1	2	3	4	6																		
Frequency	15	4	12	41	8																		
Test, at the 10% level of significance, whether or not the spinner is giving scores as it is designed to do. Show your working and state your hypotheses clearly.																							
(Total for question = 7 marks)																							
Q2. The number of hurricanes per year in a particular region was recorded over 80 years. The results are summarised in Table 1 below.																							
<table border="1" style="float: right; margin-left: auto; margin-right: auto;"> <tr> <td style="padding: 5px;">No of hurricanes, h</td> <td style="padding: 5px;">0</td> <td style="padding: 5px;">1</td> <td style="padding: 5px;">2</td> <td style="padding: 5px;">3</td> <td style="padding: 5px;">4</td> <td style="padding: 5px;">5</td> <td style="padding: 5px;">6</td> <td style="padding: 5px;">7</td> </tr> <tr> <td style="padding: 5px;">Frequency</td> <td style="padding: 5px;">0</td> <td style="padding: 5px;">2</td> <td style="padding: 5px;">5</td> <td style="padding: 5px;">17</td> <td style="padding: 5px;">20</td> <td style="padding: 5px;">12</td> <td style="padding: 5px;">12</td> <td style="padding: 5px;">12</td> </tr> </table>						No of hurricanes, h	0	1	2	3	4	5	6	7	Frequency	0	2	5	17	20	12	12	12
No of hurricanes, h	0	1	2	3	4	5	6	7															
Frequency	0	2	5	17	20	12	12	12															
(a) Write down two assumptions that will support modelling the number of hurricanes per year by a Poisson distribution.																							
(2)																							
(b) Show that the mean number of hurricanes per year from Table 1 is 4.4875																							
(2)																							
(c) Use the answer in part (b) to calculate the expected frequencies r and s given in Table 2 below to 2 decimal places.																							
<table border="1" style="float: right; margin-left: auto; margin-right: auto;"> <tr> <td style="padding: 5px;">h</td> <td style="padding: 5px;">0</td> <td style="padding: 5px;">1</td> <td style="padding: 5px;">2</td> <td style="padding: 5px;">3</td> <td style="padding: 5px;">4</td> <td style="padding: 5px;">5</td> <td style="padding: 5px;">6</td> <td style="padding: 5px;">7 or more</td> </tr> <tr> <td style="padding: 5px;">Expected frequency</td> <td style="padding: 5px;">0.90</td> <td style="padding: 5px;">4.04</td> <td style="padding: 5px;">r</td> <td style="padding: 5px;">13.55</td> <td style="padding: 5px;">s</td> <td style="padding: 5px;">13.65</td> <td style="padding: 5px;">10.21</td> <td style="padding: 5px;">13.39</td> </tr> </table>						h	0	1	2	3	4	5	6	7 or more	Expected frequency	0.90	4.04	r	13.55	s	13.65	10.21	13.39
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Expected frequency	0.90	4.04	r	13.55	s	13.65	10.21	13.39															
(3)																							
Table 2																							
(d) Test, at the 5% level of significance, whether or not the data can be modelled by a Poisson distribution. State your hypotheses clearly.																							
(6)																							
(Total 13 marks)																							

Q3. A quality control manager regularly samples 20 items from a production line and records the number of defective items x . The results of 100 such samples are given in Table 1 below.

x	0	1	2	3	4	5	6	7 or more
Frequency	17	31	19	14	9	7	3	0

Table 1

(a) Estimate the proportion of defective items from the production line.

(2)

The manager claimed that the number of defective items in a sample of 20 can be modelled by a binomial distribution. He used the answer in part (a) to calculate the expected frequencies given in Table 2.

x	0	1	2	3	4	5	6	7 or more
Expected frequency	12.2	27.0	r	19.0	s	3.2	0.9	0.2

Table 2

(b) Find the value of r and the value of s giving your answers to 1 decimal place.

(3)

(c) Stating your hypotheses clearly, use a 5% level of significance to test the manager's claim.

(7)

(d) Explain what the analysis in part (c) tells the manager about the occurrence of defective items from this production line.

(1)

(Total 13 marks)

Q4.

The number of goals scored by a football team is recorded for 100 games. The results are summarised in Table 1 below.

Number of goals	Frequency
0	40
1	33
2	14
3	8
4	5

Table 1

(a) Calculate the mean number of goals scored per game.

(2)

The manager claimed that the number of goals scored per match follows a Poisson distribution. He used the answer in part (a) to calculate the expected frequencies given in Table 2.

Number of goals	Expected Frequency
0	34.994
1	r
2	s
3	6.752
≥ 4	2.221

Table 2

(b) Find the value of r and the value of s giving your answers to 3 decimal places.

(3)

(c) Stating your hypotheses clearly, use a 5% level of significance to test the manager's claim.

(7)

(Total 12 marks)

END OF HOMEWORK

Total = 45 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.